

Training Configuration and Summary:

Sequencing:

- timesteps: 60
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

Directories:

- directory: my_dir
- project_name: lstm_hyperparameter_tuning

LSTM Training Report

Report Generated: 2025-04-09_10-03-50

Training Start: 2025-04-09 10:01:52

Training End: 2025-04-09 10:03:38

Training Duration: 105.61 seconds

Best Hyperparameters:

Number of Layers: 1

Activation: relu

Optimizer: Nadam

Dropout Rate: 0.20

Units per Layer:

Layer 1: 48 units

Performance Metrics:

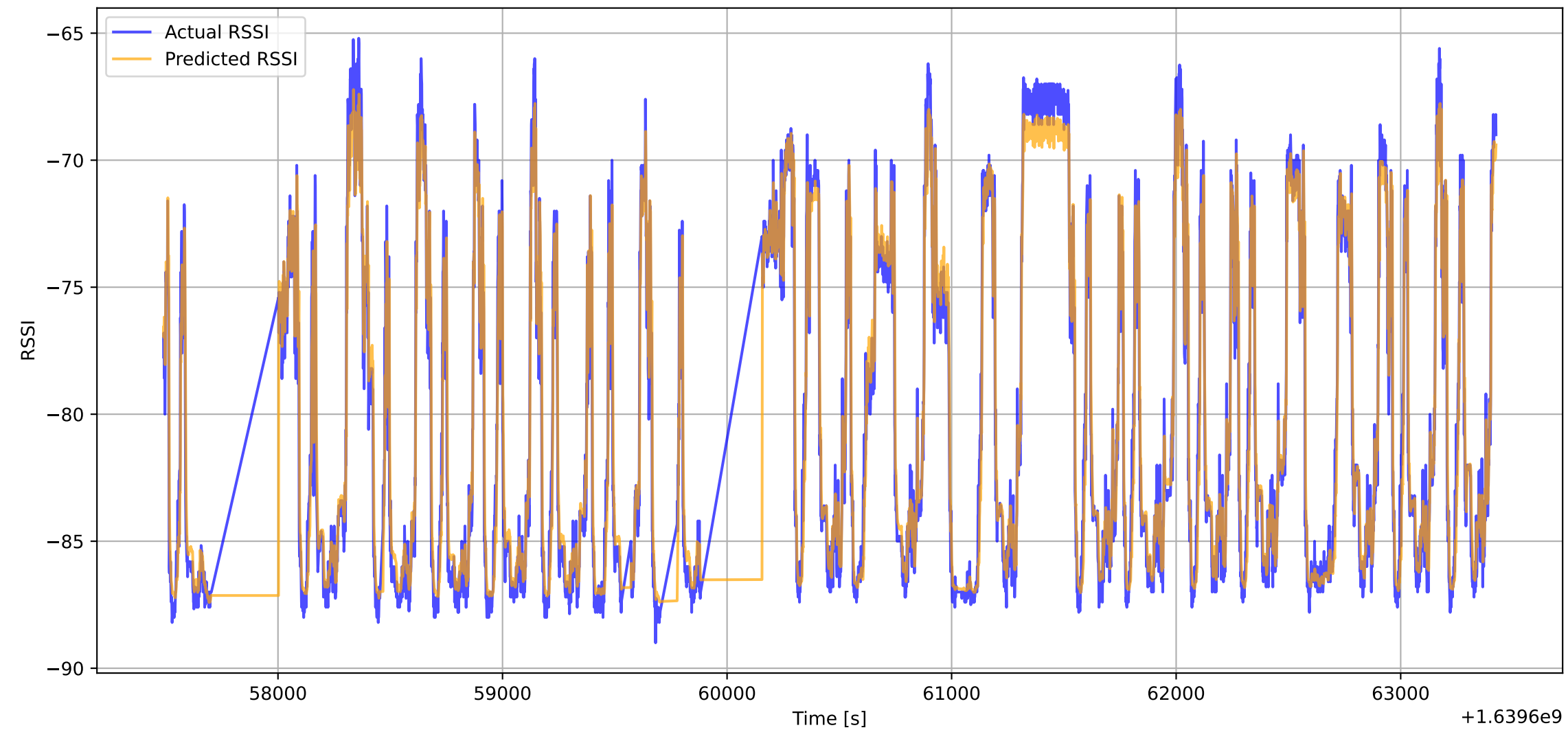
MAE: 1.0163

MSE: 2.0858

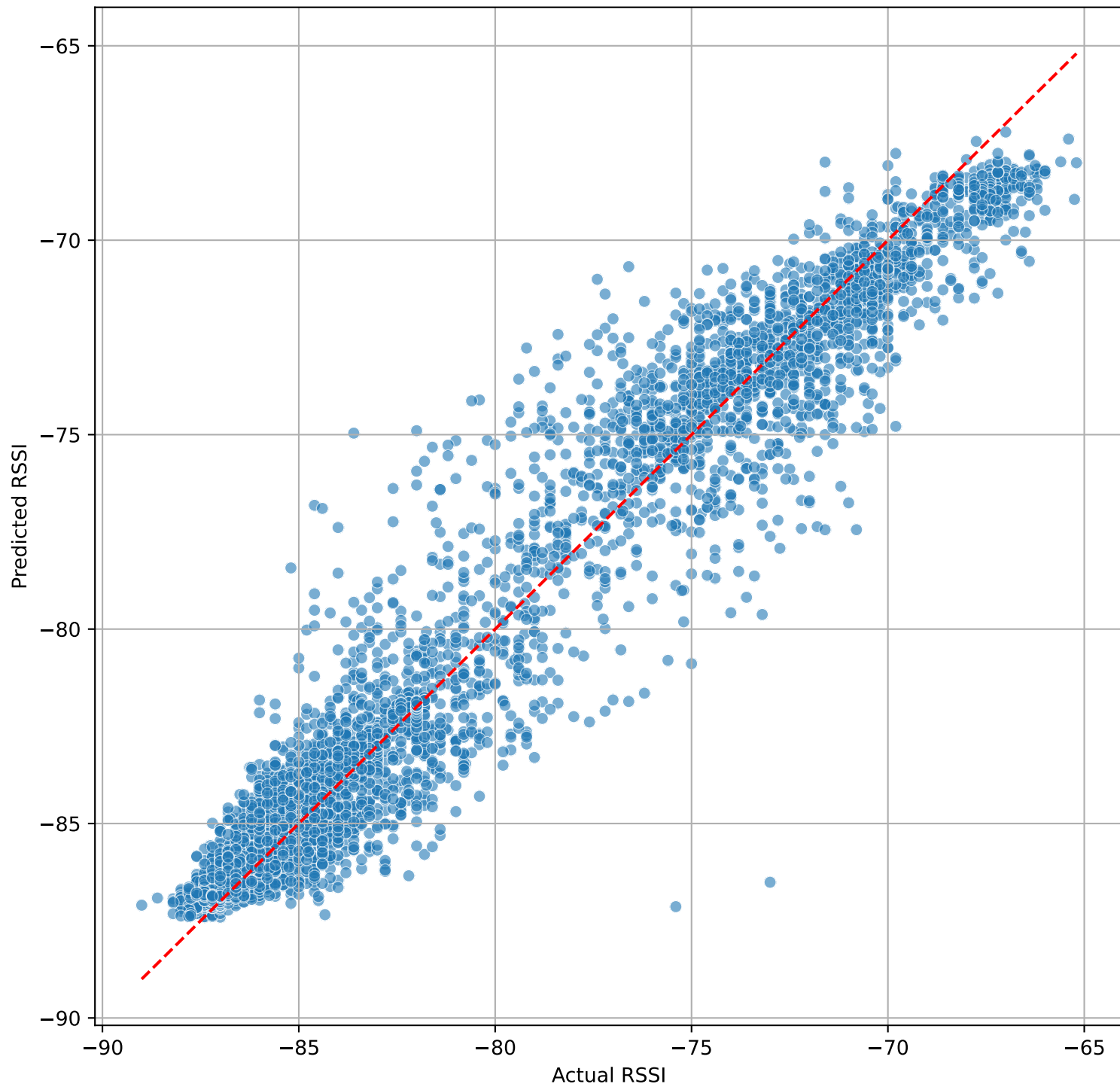
RMSE: 1.4442

R2 Score: 0.9526

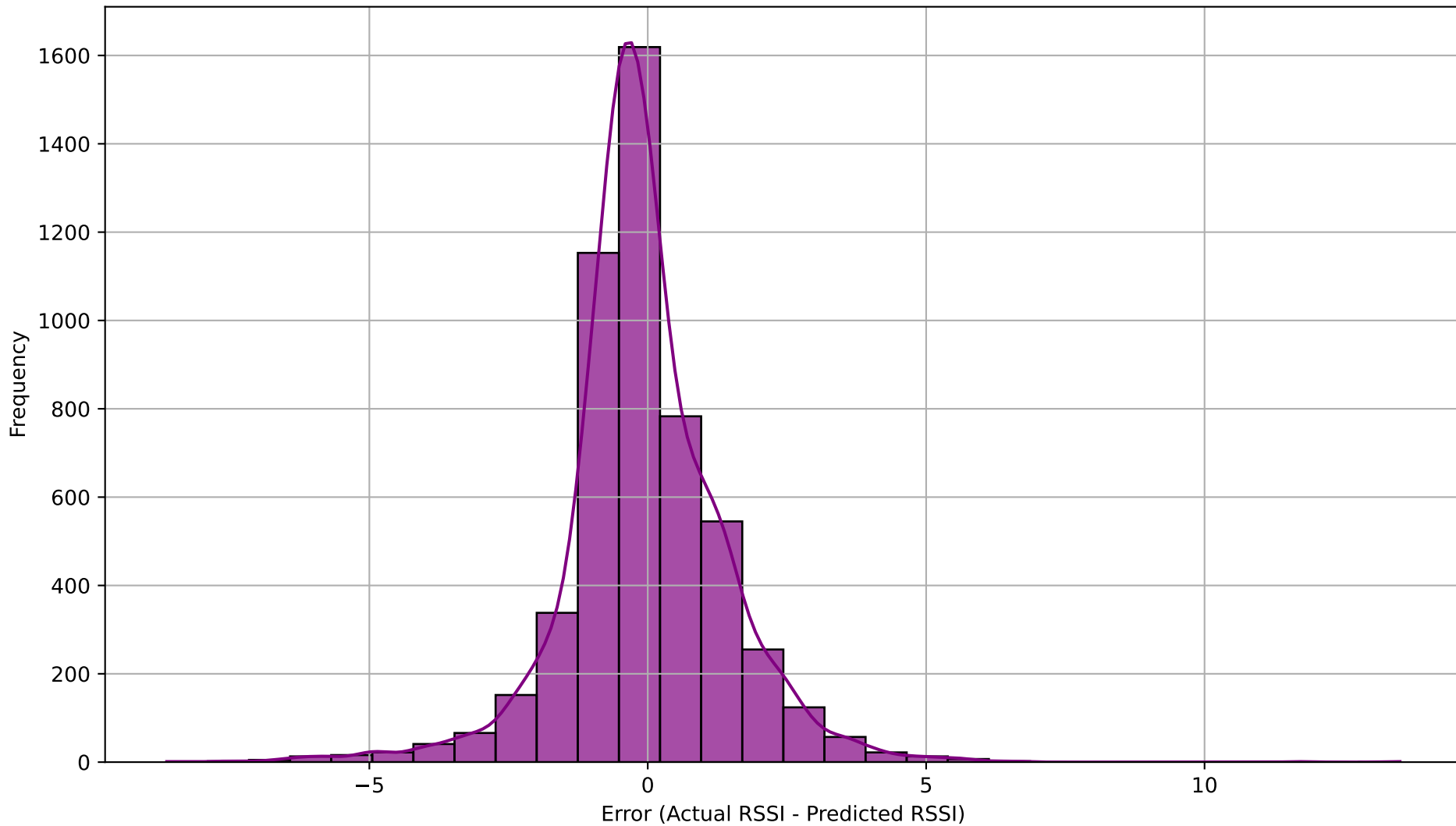
Actual vs Predicted RSSI Over Time



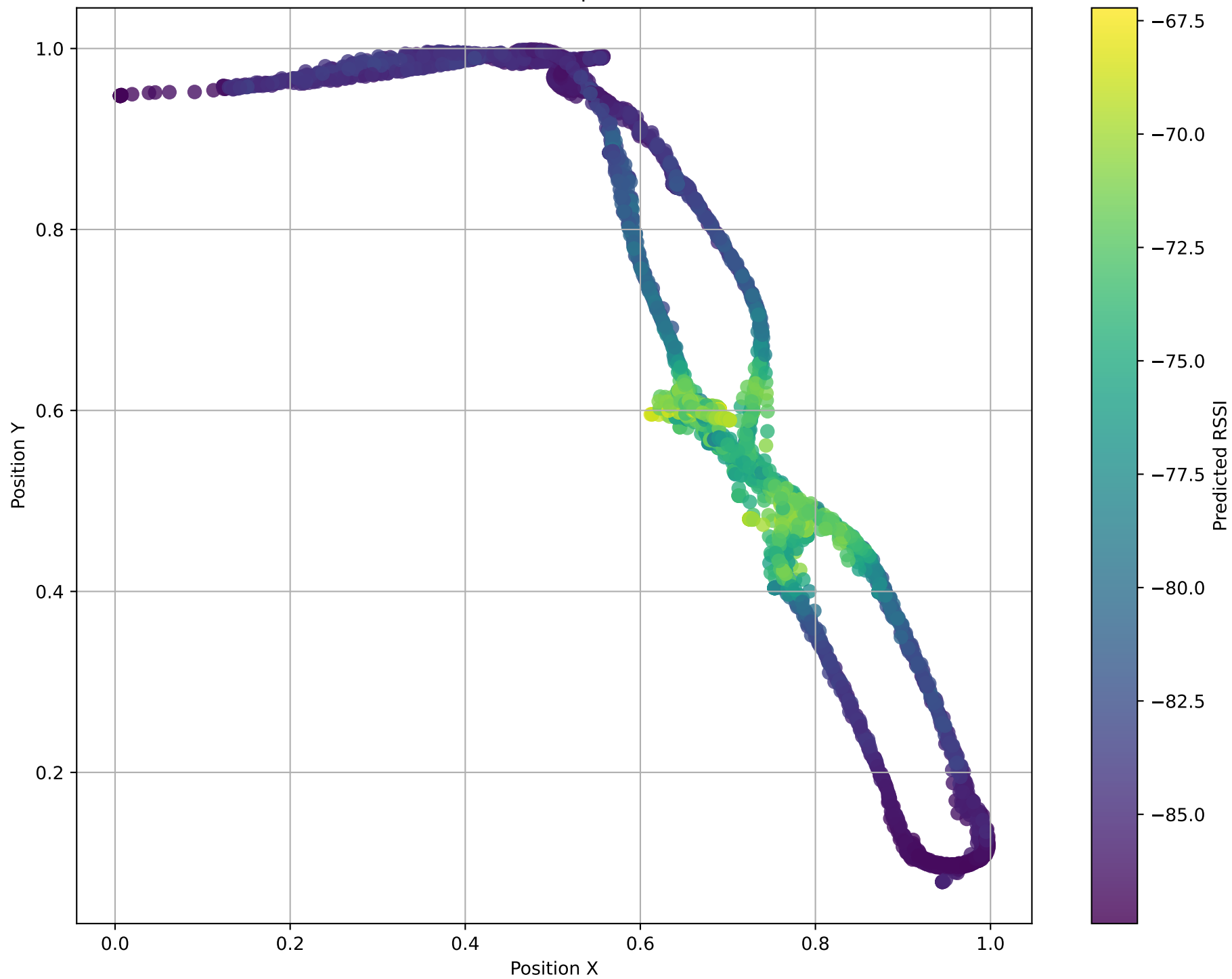
Actual vs Predicted RSSI



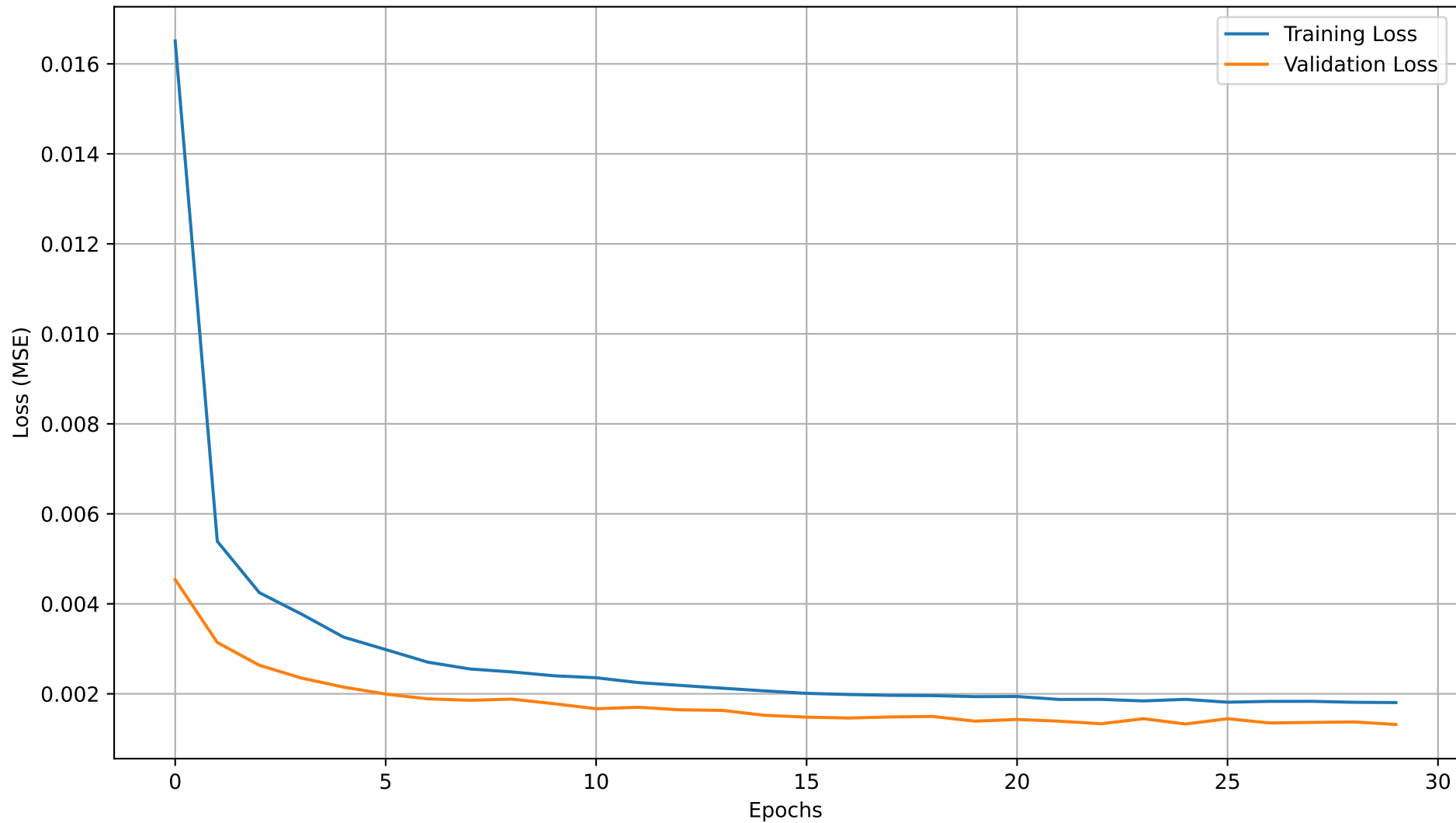
Distribution of Prediction Errors



Predicted RSSI in Spatial Context



Training and Validation Loss Over Epochs



Training and Validation MAE Over Epochs

